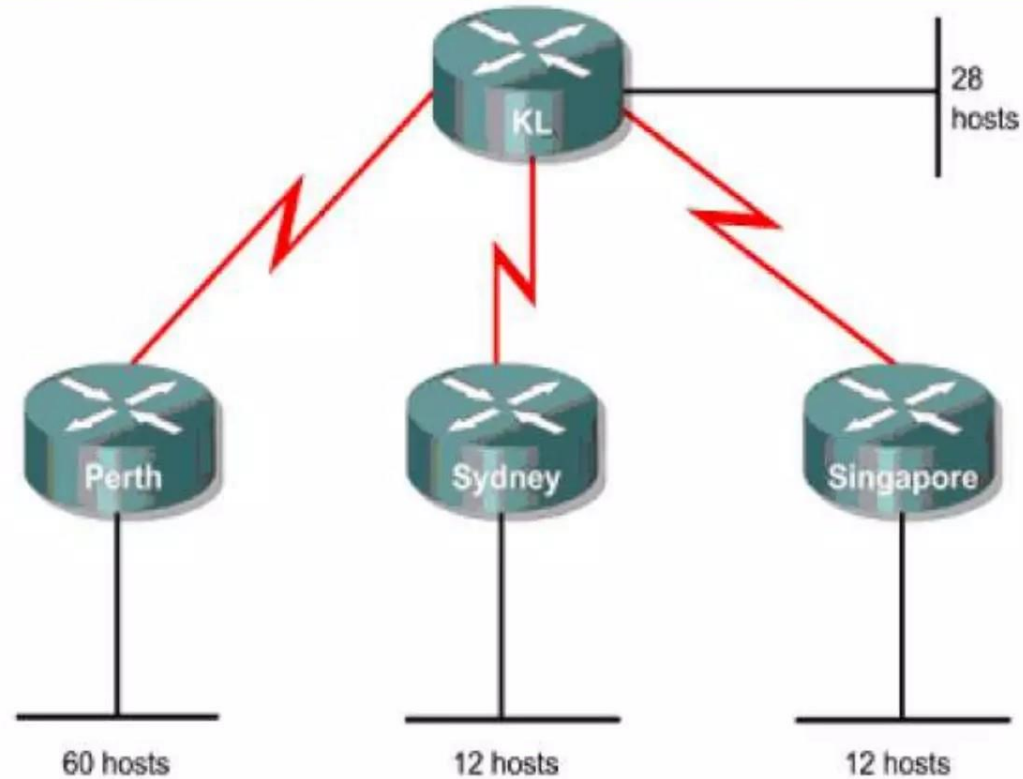


Variable Length Subnet Masking (VLSM)

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VLSM (By Example)



- Class C address 192.168.10.0/24 has been allocated
- Perth, Sydney and Singapore have a WAN connection to Kuala Lumpur
- Hosts Required:
 - Perth – 60, Kuala Lumpur – 28, Sydney – 12, Singapore – 12 hosts

VLSM (By Example)

- To calculate VLSM subnets and the respective hosts allocate the **largest requirements first** from the address range
- Requirements levels should be listed from the largest to the smallest
- In this example, Perth requires 60 hosts (largest)
 - Use 6 bits, since $2^6 - 2 = 62$ usable addresses
 - 2 bits will be used from the 4th octet to represent the extended-network-prefix of **/26**
 - Remaining 6 bits will be used for host addresses

VLSM (By Example)

Step I : Divide the allocated address of 192.168.10.0/24 into four equal-size address blocks

- Since $4 = 2^2$, 2 bits are required to identify each of the 4 subnets
- Next take subnet #0 (192.168.10.0/26) and identify its hosts

Allocated Address	Sub-networks	62 usable addresses/sub-network (subnet #0)
192.168.10.0/24	192.168.10.0/26	192.168.10.0/26 (Network Address)
	192.168.10.64/26	192.168.10.1/26 (for Router interface)
	192.168.10.128/26	192.168.10.2/26 (Host 1)
	192.168.10.192/26	192.168.10.3/26 (Host 2)
		thru
		192.168.10.61/26 (Host 60)
		192.168.10.62/26 (unused)
		192.168.10.63/26 (Broadcast Address)

VLSM (By Example)

Here is the range for /26 mask

Perth	Range of addresses in the last octet
192.168.10.0/26	From 0 to 63, 60 hosts are required Hosts 0 and 63 cannot be used because they are the network and broadcast addresses for their subnet First usable address given to router interface Next 60 addresses given to hosts Last address is unused

VLSM (By Example)

Step II: Allocate next level after all requirements for higher level

- Kuala Lumpur requires 28 hosts; Next available address is 192.168.10.64/26
- 5 bits needed for hosts, since $2^5 - 2 = 30$ usable addresses
- 3 bits will be used to represent the extended-network-prefix of /27
- Applying VLSM on 192.168.10.64/26

Sub-network #1	Sub-sub-networks	30 usable addresses
192.168.10.64/26	192.168.10.64/27	192.168.10.64/27 (Network Address)
	192.168.10.96/27	192.168.10.65/27 (for Router interface)
		192.168.10.66/27 (Host 1)
		192.168.10.67/27 (Host 2)
		thru
		192.168.10.93/27 (Host 28)
		192.168.10.94/27 (unused)
		192.168.10.95/27 (Broadcast Address)

VLSM (By Example)

Here is the range for /27 mask

Kuala Lumpur	Range of addresses in the last octet
192.168.10.64/27	From 64 to 95, 28 hosts are required Hosts 64 and 95 cannot be used because they are the network and broadcast addresses for their subnet First usable address given to router interface Next 28 addresses given to hosts Last address is unused

VLSM (By Example)

Step III : Allocate next level after all requirements for higher level

- Sydney requires 12 hosts; Next available address is 192.168.10.96/27
- 4 bits needed for hosts, since $2^4 - 2 = 14$ usable addresses
- 4 bits will be used to represent the extended-network-prefix of /28
- Applying VLSM on 192.168.10.96/27

Sub-network #1	Sub-sub-networks	14 usable addresses
192.168.10.96/27	192.168.10.96/28	192.168.10.96/28 (Network Address)
	192.168.10.112/28	192.168.10.97/28 (for Router interface)
		192.168.10.98/28 (Host 1)
		192.168.10.99/28 (Host 2)
		thru
		192.168.10.109/28 (Host 12)
		192.168.10.110/28 (unused)
		192.168.10.111/28 (Broadcast Address)

VLSM (By Example)

Here is the range for /28 mask

Sydney	Range of addresses in the last octet
192.168.10.96/28	From 96 to 111, 12 hosts are required Hosts 96 and 111 cannot be used because they are the network and broadcast addresses for their subnet First usable address given to router interface Next 12 addresses given to hosts Last address is unused

VLSM (By Example)

Step IV : Allocate next level after all requirements for higher level

- Singapore requires 12 hosts; Next available address is 192.168.10.112/28
- 4 bits needed for hosts, since $2^4 - 2 = 14$ usable addresses
- 4 bits will be used to represent the extended-network-prefix of /28
- Using 192.168.10.112/28

Sub-network #1	Sub-sub-networks	14 usable addresses
192.168.10.96/27	192.168.10.112/28	192.168.10.112/28 (Network Address)
		192.168.10.113/28 (for Router interface)
		192.168.10.114/28 (Host 1)
		192.168.10.115/28 (Host 2)
		Thru
		192.168.10.125/28 (Host 12)
		192.168.10.126/28 (unused)
		192.168.10.127/28 (Broadcast Address)

VLSM (By Example)

Here is the range for /28 mask

Singapore	Range of addresses in the last octet
192.168.10.112/28	From 112 to 127, 12 hosts are required Hosts 112 and 127 cannot be used because they are the network and broadcast addresses for their subnet First usable address given to router interface Next 12 addresses given to hosts Last address is unused

VLSM (By Example)

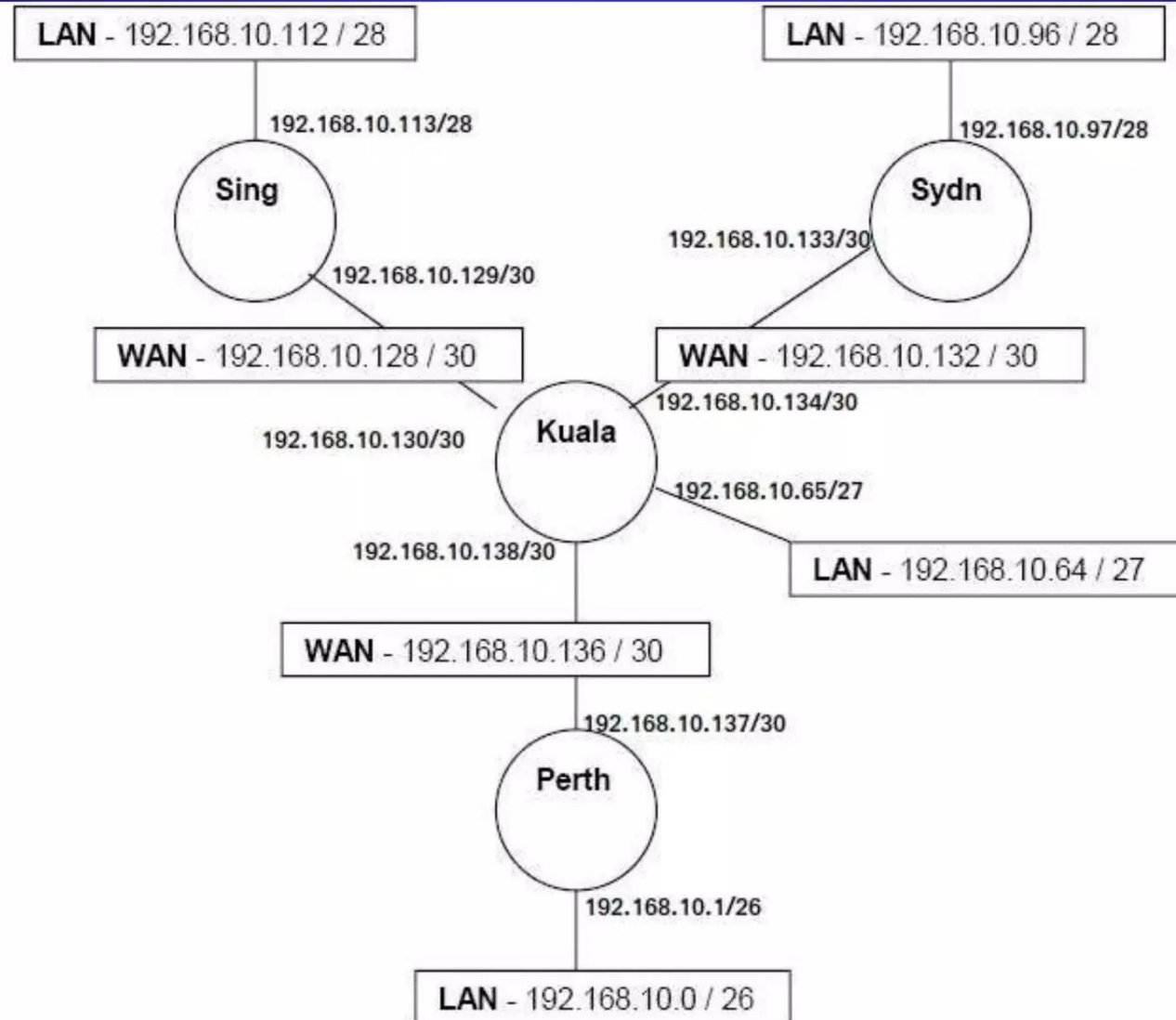
Step V : Now allocate addresses for WAN links

- Each WAN link requires 2 IP addresses
- Next available address is 192.168.10.128/28
- 2 bits needed, since $2^2 - 2 = 2$ usable addresses
- 2 bits required to represent the links
- 6 bits for the extended-network-prefix of /30
- Applying VLSM on 192.168.10.112/28

VLSM (By Example)

Sub-sub-networks	14 usable addresses
192.168.10.128/30	192.168.10.128/30 (Network Address)
(WAN Link A b/w Kuala Lumpur and Singapore)	192.168.10.129/30 (interface 1 at Singapore)
	192.168.10.130/30 (interface 2 at Kuala Lumpur)
	192.168.10.131/30 (Broadcast Address)
192.168.10.132/30	192.168.10.132/30 (Network Address)
(WAN Link B b/w Kuala Lumpur and Sydney)	192.168.10.133/30 (interface 1 at Sydney)
	192.168.10.134/30 (interface 2 at Kuala Lumpur)
	192.168.10.135/30 (Broadcast Address)
192.168.10.136/30	192.168.10.136/30 (Network Address)
(WAN Link C b/w Kuala Lumpur and Perth)	192.168.10.137/30 (interface 1 at Perth)
	192.168.10.138/30 (interface 2 at Kuala Lumpur)
	192.168.10.139/30 (Broadcast Address)

VLSM (By Example) – Solution Diagram



VLSM (By Example)

- This is how the Class C address was subnetted using VLSM
- The network address and broadcast address for each subnet are shown in the corners of each box

